GATE ME 90 Days Challenge

Day 63/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	**Probability & Statistics**	**38 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on probability rules, independence and conditional probability, Bayes' theorem, discrete and continuous random variables, expectation and variance, binomial, Poisson, exponential, uniform and normal distributions, mean, median, mode and standard deviation, correlation and least-squares regression. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A and B are two events with $P(A) > 0$ and $P(B) > 0$. If A and B are mutually exclusive, then

(A) A and B must also be independent
(B) A and B cannot be independent
(C) $P(A \cup B) = P(A)P(B)$
(D) $P(A \mid B) = P(A)$

Q2 NAT

A machine has four critical components that fail independently, each with a probability of 0.05 during a shift. The machine stops if any one of them fails. The probability that the machine stops during the shift (round off to three decimal places) is _____

Q3 MCQ

For which one of the following distributions are the mean and the variance always equal?

(A) Binomial
(B) Normal
(C) Poisson
(D) Uniform

Q4 NAT

30 % of the e-mails reaching an inbox are spam. The word 'offer' appears in 40 % of spam e-mails and in 5 % of genuine e-mails. An e-mail is found to contain 'offer'. The probability that it is spam (round off to three decimal places) is _____

Q5 MCQ

X is normally distributed with mean 10 and variance 4. Given $\Phi(1) = 0.8413$, where Φ is the standard normal cumulative distribution function, $P(8 < X < 12)$ is

(A) 0.1587
(B) 0.6826
(C) 0.8413
(D) 0.3413

Q6 NAT

A shop has 6 identical machines, each of which is independently under repair on a given day with a probability of 0.2. The probability that exactly 2 machines are under repair on a given day (round off to three decimal places) is _____

Q7 MCQ

A random variable X has the probability density function $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ ($\lambda > 0$). Its mean and variance are

(A) $1/\lambda$ and $1/\lambda^2$
(B) λ and λ
(C) $1/\lambda$ and $1/\lambda$
(D) λ and λ^2

Q8 NAT

A random variable X has the probability density function $f(x) = k(1 - x^2)$ for $-1 \leq x \leq 1$ and zero elsewhere. The value of k (round off to two decimal places) is _____

Q9 MCQ

The mean of 10 observations was found to be 20. Later it was noticed that one observation, 38, had been wrongly recorded as 83. The correct mean is

(A) 24.5
(B) 20.0
(C) 15.5
(D) 11.0

Q10 NAT

The standard deviation (population, dividing by n) of the data 4, 8, 6, 10, 12 is _____ (round off to two decimal places)

Q11 MCQ

X takes the values $-2, -1, 0, 1, 2$ with equal probability and $Y = X^2$. The correlation coefficient between X and Y is

(A) 1, because Y is completely determined by X
(B) -1
(C) 0.5
(D) 0, although X and Y are not independent

Q12 NAT

For 5 pairs of observations, $\Sigma x = 20$, $\Sigma y = 30$, $\Sigma xy = 140$ and $\Sigma x^2 = 100$. Using the least-squares regression line of y on x, the estimated value of y at $x = 6$ is _____

Q13 MCQ

A process produces 2 % defective items. Using the Poisson approximation, the probability that a lot of 100 items contains no defective item is about

(A) 0.135
(B) 0.020
(C) 0.271
(D) 0.865

Q14 NAT

A random variable has the probability density function $f(x) = 3x^2$ for $0 \leq x \leq 1$ and zero elsewhere. Its variance (round off to four decimal places) is _____

Q15 MCQ

The diameters of pins have a standard deviation of 12 μm. Samples of 36 pins are taken and the sample mean is calculated. The standard deviation of the sample mean (standard error) is

(A) 12 μm
(B) 0.33 μm
(C) 6 μm
(D) 2 μm

Answer Key

Q	Type	Answer
Q1	MCQ	(B) A and B cannot be independent
Q2	NAT	0.185 (accept 0.184 to 0.187)
Q3	MCQ	(C) Poisson
Q4	NAT	0.774 (accept 0.770 to 0.778)
Q5	MCQ	(B) 0.6826
Q6	NAT	0.246 (accept 0.244 to 0.248)
Q7	MCQ	(A) $1/\lambda$ and $1/\lambda^2$
Q8	NAT	0.75
Q9	MCQ	(C) 15.5
Q10	NAT	2.83 (accept 2.81 to 2.85)
Q11	MCQ	(D) 0, although X and Y are not independent
Q12	NAT	8
Q13	MCQ	(A) 0.135
Q14	NAT	0.0375
Q15	MCQ	(D) 2 µm

Detailed Solutions

Q1 **Answer: (B) A and B cannot be independent**

Concept: Mutually exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A)P(B)$.

Since $P(A)$ and $P(B)$ are both positive, $P(A)P(B) > 0 \neq P(A \cap B) = 0$, so they cannot be independent: **(B)**.

Indeed $P(A \mid B) = 0$: knowing B happened tells us for sure that A did not, which is the opposite of independence.

Exam tip: 'Mutually exclusive' is about the events' outcomes; 'independent' is about probabilities. Do not mix them up.

Q2 **Answer: 0.185 (accept 0.184 to 0.187)**

Concept: P(at least one) = 1 − P(none).

P(no failure) = $0.95^4 = 0.81451$

P(stop) = 1 − 0.81451 = **0.185**

Exam tip: Simply adding 4 × 0.05 = 0.20 overcounts the cases where two or more fail.

Q3 **Answer: (C) Poisson**

Concept: For a Poisson distribution with parameter λ, E[X] = Var(X) = λ.

A binomial has variance np(1 − p) < np; a normal or uniform distribution has mean and variance set independently: **(C)**.

Exam tip: If the sample mean and variance of count data are nearly equal, a Poisson model is a good first guess.

Q4 **Answer: 0.774 (accept 0.770 to 0.778)**

Concept: Bayes' theorem: P(S | W) = P(W | S)P(S)/[P(W | S)P(S) + P(W | G)P(G)].

Numerator = 0.4 × 0.3 = 0.12; denominator = 0.12 + 0.05 × 0.7 = 0.155

P(S | W) = 0.12/0.155 = **0.774**

Exam tip: Draw a tree: the two branches that end in 'offer' form the denominator.

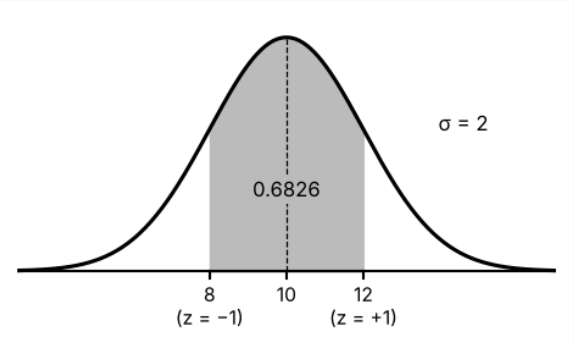
Q5 **Answer: (B) 0.6826**

Concept: Standardise with z = (x − μ)/σ; note that σ = √4 = 2, not 4.

$z_1 = (8 - 10)/2 = -1$, $z_2 = (12 - 10)/2 = 1$

P = Φ(1) − Φ(−1) = 2Φ(1) − 1 = 2 × 0.8413 − 1 = **0.6826 (B)**

Exam tip: About 68 %, 95 % and 99.7 % of a normal population lie within 1, 2 and 3 standard deviations of the mean.



Q6 **Answer: 0.246 (accept 0.244 to 0.248)**

Concept: Binomial: $P(X = r) = {}^{n}C_{r}\, p^{r}(1 - p)^{n-r}$.

$P(X = 2) = 15 \times 0.2^{2} \times 0.8^{4} = 15 \times 0.04 \times 0.4096$

P = **0.246**

Exam tip: Check that the trials are independent and p is the same for all; otherwise the binomial model does not apply.

Q7 **Answer: (A) 1/λ and 1/λ²**

Concept: For the exponential distribution, $E[X] = \int x\lambda e^{-\lambda x}\, dx = 1/\lambda$ and $E[X^2] = 2/\lambda^2$.

$Var(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$, so the mean and the standard deviation are equal: **(A)**.

Exam tip: λ is a rate (failures per hour); 1/λ is the mean time between failures. (λ, λ) belongs to the Poisson distribution.

Q8 **Answer: 0.75**

Concept: The total area under a pdf is 1.

$\int$ from -1 to 1 of $(1 - x^2)\,dx = [x - x^3/3]$ from -1 to $1 = 2/3 + 2/3 = 4/3$

$k \times 4/3 = 1$, so $k =$ **0.75**

Exam tip: Also check $f(x) \geq 0$ everywhere in the range; a 'pdf' that goes negative is not valid whatever k is.

Q9 **Answer: (C) 15.5**

Concept: Correct the sum, not the mean.

Wrong sum $= 10 \times 20 = 200$. Correct sum $= 200 - 83 + 38 = 155$

Correct mean $= 155/10 =$ **15.5**

Exam tip: A single wrong entry shifts the mean by (error)/n $= 45/10 = 4.5$. The median is far less sensitive to such outliers.

Q10 **Answer: 2.83 (accept 2.81 to 2.85)**

Concept: $\sigma = \sqrt{[\Sigma(x - \bar{x})^2/n]}$.

$\bar{x} = 40/5 = 8$; squared deviations: 16, 0, 4, 4, 16, total 40

$\sigma^2 = 40/5 = 8$, $\sigma = \sqrt{8} =$ **2.83**

Exam tip: The sample standard deviation (divide by $n - 1$) would be $\sqrt{10} = 3.16$; read which one the question wants.

Q11 **Answer: (D) 0, although X and Y are not independent**

Concept: The correlation coefficient measures only *linear* association: $r \propto \text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.

$E[X] = 0$ and $E[XY] = E[X^3] = (-8 - 1 + 0 + 1 + 8)/5 = 0$, so Cov $= 0$ and $r = 0$: **(D)**.

Yet Y is a function of X, so they are clearly dependent.

Exam tip: Independent $\Rightarrow$ uncorrelated, but uncorrelated does not imply independent (except for jointly normal variables).

Q12 **Answer: 8**

Concept: $b_{yx} = [\Sigma xy - (\Sigma x)(\Sigma y)/n]/[\Sigma x^2 - (\Sigma x)^2/n]$, and the line passes through $(\bar{x}, \bar{y})$.

$b_{yx} = (140 - 120)/(100 - 80) = 1$; $\bar{x} = 4$, $\bar{y} = 6$

$y = 6 + 1 \times (6 - 4) =$ **8**

Exam tip: Use the y-on-x line to predict y; the x-on-y line is a different line unless $r = \pm 1$.

Q13 **Answer: (A) 0.135**

Concept: For large n and small p, the binomial is approximated by a Poisson distribution with $\lambda = np$.

$\lambda = 100 \times 0.02 = 2$, so $P(0) = e^{-2} =$ **0.135 (A)**

The exact binomial value $0.98^{100} = 0.133$ is very close.

Exam tip: 0.271 is $P(X = 1) = 2e^{-2}$, a common slip.

Q14 **Answer: 0.0375**

Concept: $\text{Var}(X) = E[X^2] - (E[X])^2$.

$E[X] = \int 3x^3\,dx = 3/4$; $E[X^2] = \int 3x^4\,dx = 3/5$

$\text{Var} = 0.6 - 0.5625 = \mathbf{0.0375}$

Exam tip: For $f(x) = (n + 1)x^n$ on [0, 1], $E[X] = (n + 1)/(n + 2)$, a handy check.

Q15 **Answer: (D) 2 µm**

Concept: For n independent observations, $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.

$\sigma_{\bar{x}} = 12/\sqrt{36} = 12/6 = \mathbf{2\ \mu m\ (D)}$

By the central limit theorem the sample mean is nearly normal even if the individual diameters are not.

Exam tip: This is why $\bar{X}$ control charts have limits at $\mu \pm 3\sigma/\sqrt{n}$, much narrower than the spread of individual parts.

What next?

Day 63 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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